Chapter 13

Digital Signature

Chapter 13

Objectives

- To define a digital signature
- To define security services provided by a digital signature
- To define attacks on digital signatures
- To discuss some digital signature schemes, including RSA, ElGamal,
- Schnorr, DSS, and elliptic curve
- To describe some applications of digital signatures

13-1 COMPARISON

Let us begin by looking at the differences between conventional signatures and digital signatures.

Topics discussed in this section:

13.1.1 Inclusion 390
13.1.2 Verification Method 390
13.1.3 Relationship 390
13.1.4 Duplicity 390

13.1.1 Inclusion

A conventional signature is included in the document; it is part of the document.

But when we sign a document digitally, we send the signature as a separate document.

13.1.2 Verification Method

For a conventional signature, when the recipient receives a document, she compares the signature on the document with the signature on file.

:A Cheque

For a digital signature, the recipient receives the message and the signature. The recipient needs to apply a verification technique to the combination of the message and the signature to verify the authenticity.

13.1.3 Relationship

For a conventional signature, there is normally a one-to-many relationship between a signature and documents.

For a digital signature, there is a one-to-one relationship between a signature and a message.

13.1.4 Duplicity

In conventional signature, a copy of the signed document can be distinguished from the original one on file.

In digital signature, there is no such distinction unless there is a factor of time on the document.

Need of Digital Signature

Message authentication protects the two parties from any third party. However, it does not protect the two parties against each other.

Disputes:

- *B may forge a different document and and claim that it is from A.*
  - *Ex- funds transfer.*
- *A can deny, sending the message.*
  - *Ex- instruction to a stock-broker.*

Non-repudiation: An entity should not be allowed to deny valid signatures made by him.

Desired properties of digital Signature

- *It must verify the author and the date and time of the signature.*
- *It must authenticate the contents at the time of signature.*
- *It must be verifiable by a third party, to restore disputes.*

direct and arbitrated digital signature

Direct digital signature	Arbitrated Digital Signature
1.The direct digital signature involves only the communicating parties. 2.This may be formed by encrypting the entire message with the sender's private key.	1.The arbiter plays a sensitive and crucial role in this digital signature. 2. Every signed message from a sender x to a receiver y goes first to an arbiter A, who subjects the message and its signature to a number of tests to check its origin and content.

13-2 PROCESS

Figure 13.1 shows the digital signature process.
The sender uses a signing algorithm to sign the message.
The message and the signature are sent to the receiver.
The receiver receives the message and the signature and applies the verifying algorithm to the combination.

If the result is true, the message is accepted; otherwise, it is rejected.

Topics discussed in this section:

13.2.1 Need for Keys
13.2.2 Signing the Digest

Process

Alice establishes her binding to a message M by digitally signing it.

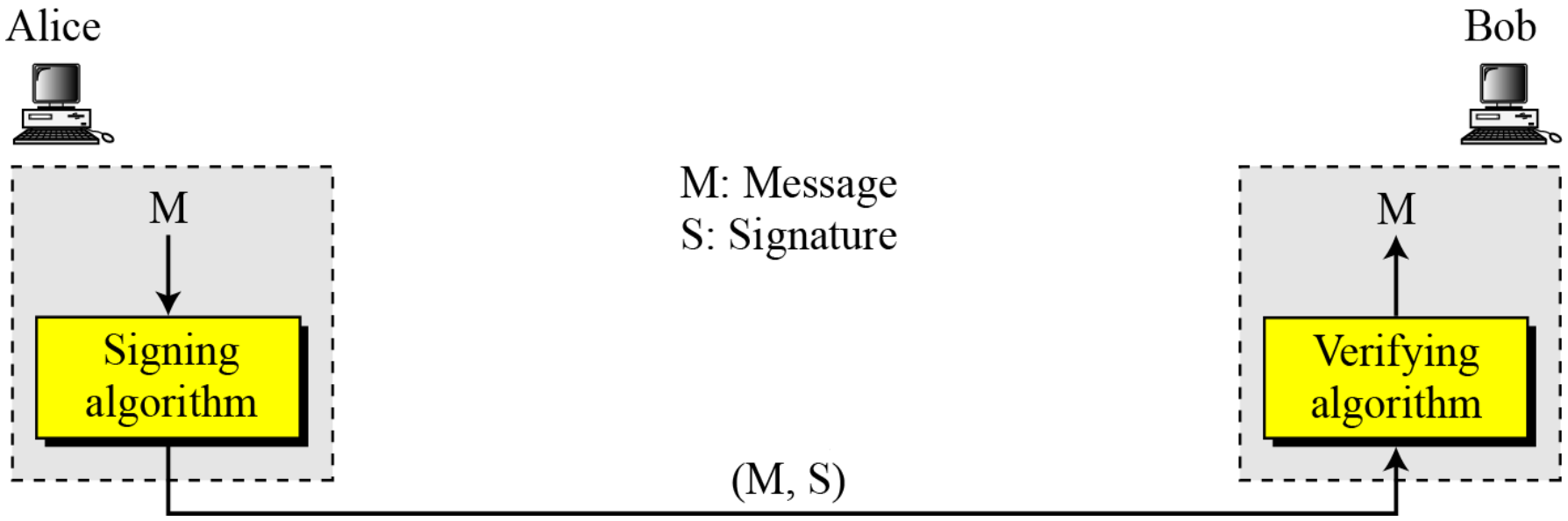
- *Signing: Only Alice has the capability to sign M.*
- *Verification: Anybody can verify whether Alice's signature on M is valid.*
- *Forging: Nobody can forge signatures on behalf of Alice.*

Digital signatures are based on public-key techniques.

- *Signature generation ≡ Decryption (uses private key), and*
- *Signature verification ≡ Encryption (uses public key).*

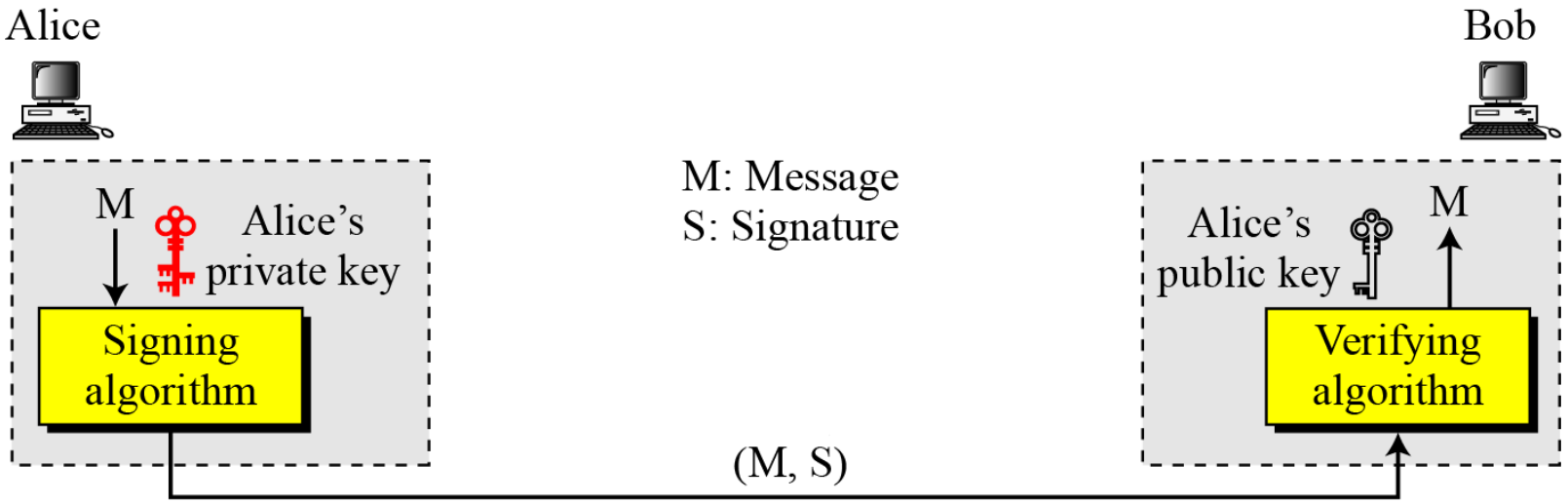
13-2 Continued

Figure 13.1 *Digital signature process*



13.2.1 Need for Keys

Figure 13.2 *Adding key to the digital signature process*



Note

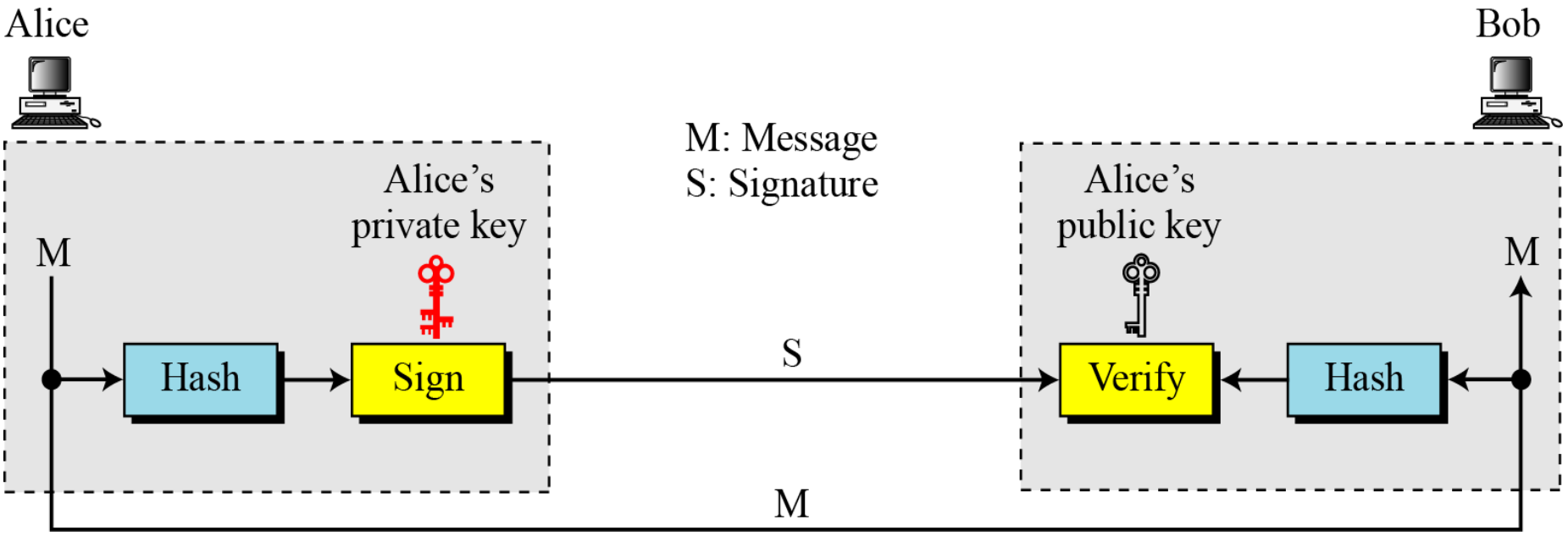
A digital signature needs a public-key system. The signer signs with her private key; the verifier verifies with the signer's public key.

Note

A cryptosystem uses the private and public keys of the receiver: a digital signature uses the private and public keys of the sender.

13.2.2 Signing the Digest

Figure 13.3 *Signing the digest*



13-3 SERVICES

We discussed several security services in Chapter 1 including message confidentiality, message authentication, message integrity, and nonrepudiation.

A digital signature can directly provide the last three; for message confidentiality we still need encryption/decryption.

Topics discussed in this section:

13.3.1 Message Authentication
13.3.2 Message Integrity
13.3.3 Nonrepudiation
13.3.4 Confidentiality

13.3.1 Message Authentication

A secure digital signature scheme, like a secure conventional signature can provide message authentication.

Note

A digital signature provides message authentication.

13.3.1 Message Authentication Requirements

i. Disclosure: Release of message contents to any person or process not processing the appropriate cryptographic key
ii. Traffic Analysis: Discovery of the pattern of traffic between parties. In a connection oriented application, the frequency and duration of connections could be determined. In either a connection oriented or connectionless environment, the number and length of messages between parties could be determined.
iii. Masquerade: Insertion of messages into the network from a fraudulent source. This includes the creation of messages by an opponent that are purported to come from an authorized entity. Also included are fraudulent acknowledgements of message receipt or no receipt by someone other than the message recipient.
iv. Content modification: Changes to the contents of a message , including insertion, deletion, transposition, and modification.
v. Sequence modification: Any modification to a sequence of messages between parties, including insertion, deletion, and modification.
vi. Timing modification: Delay or replay of messages. In a connection oriented application, an entire session or sequence of messages could be a replay of some previous valid session, or individual messages in the sequence could be delayed or replayed. In connectionless application, an individual message could be delayed or replayed.
vii. Source repudiation: Denial of transmission of message by source. viii. Destination repudiation: Denial of receipt of message by destination.

13.3.2 Message Integrity

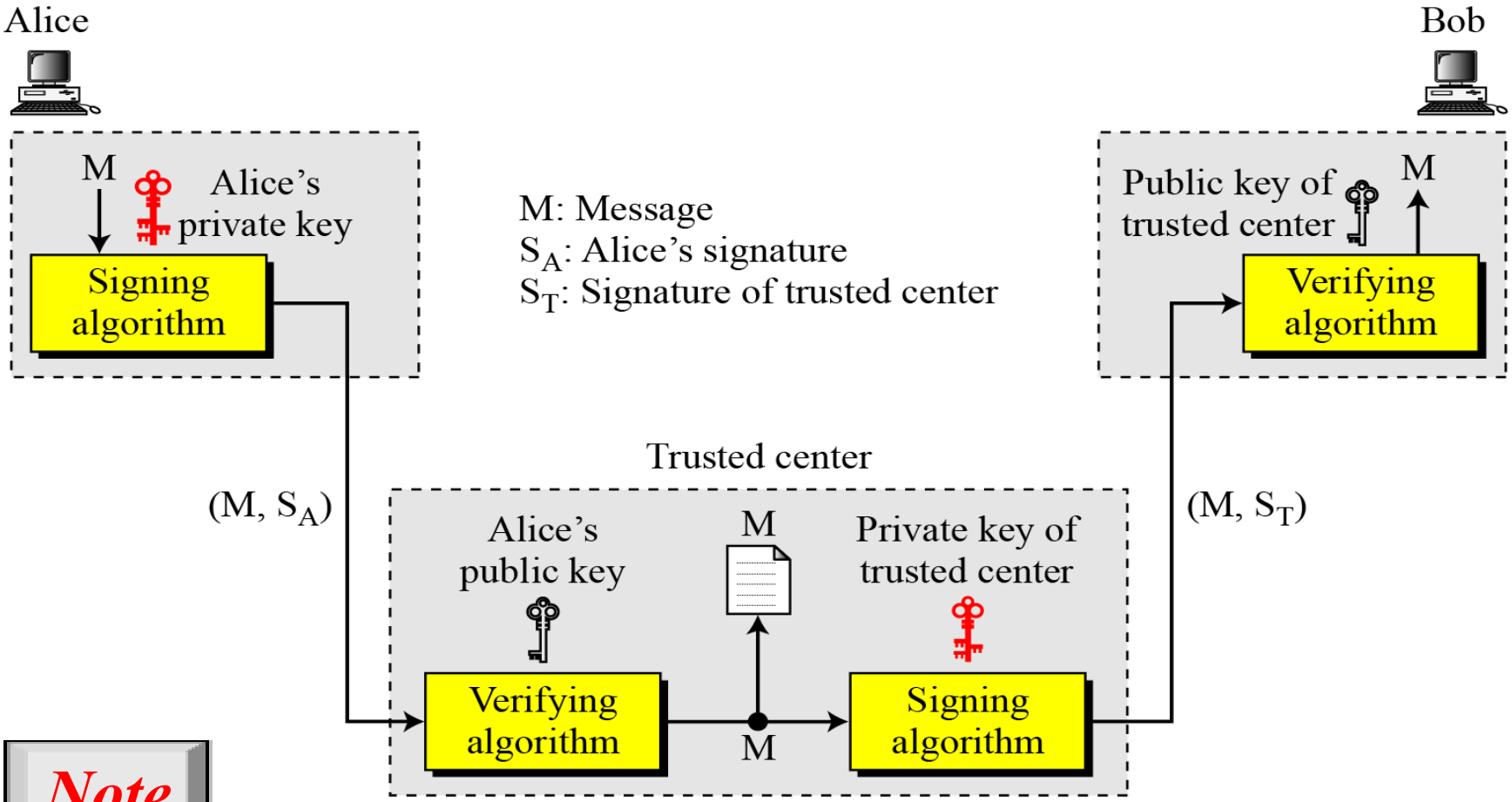
The integrity of the message is preserved even if we sign the whole message because we cannot get the same signature if the message is changed.

Note

A digital signature provides message integrity.

13.3.3 Nonrepudiation

Figure 13.4 *Using a trusted center for nonrepudiation*

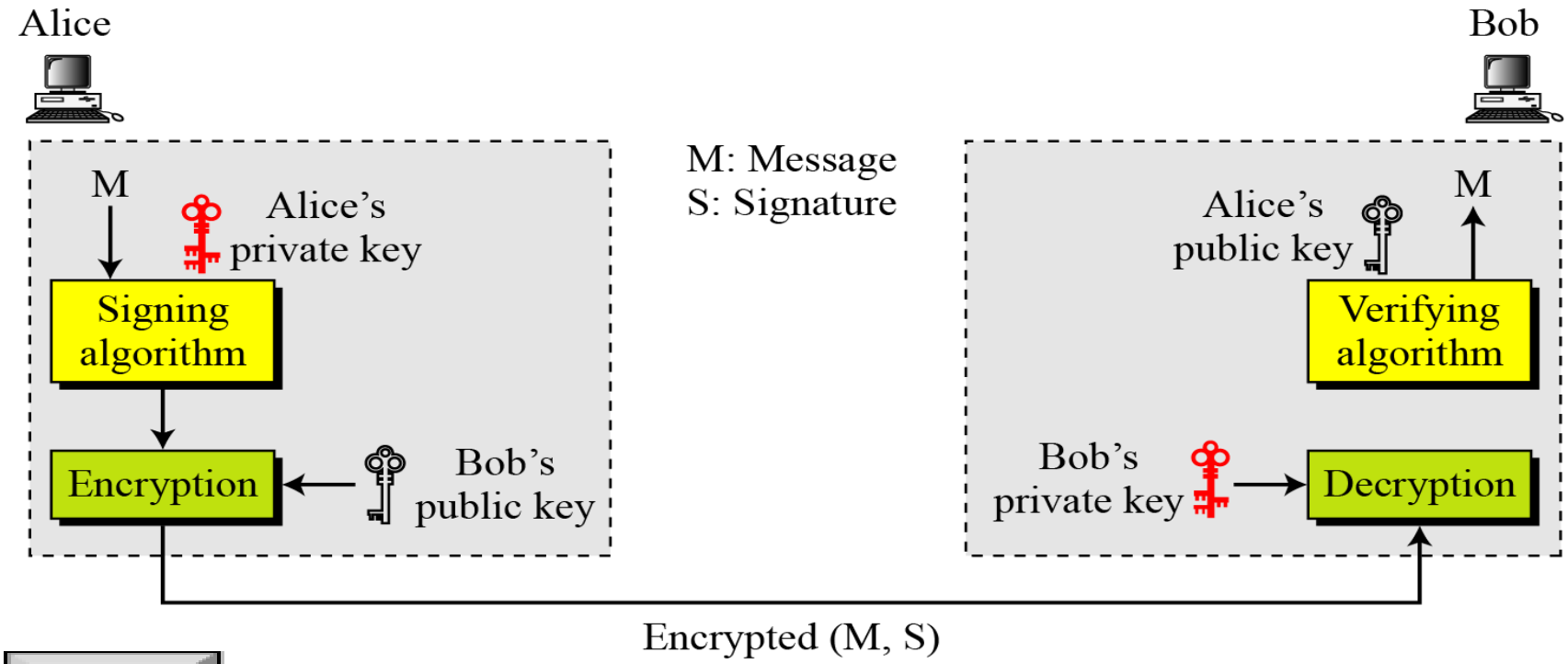


Note

Nonrepudiation can be provided using a trusted party.

13.3.4 Confidentiality

Figure 13.5 *Adding confidentiality to a digital signature scheme*



Note

A digital signature does not provide privacy. If there is a need for privacy, another layer of encryption/decryption must be applied.

13-5 DIGITAL SIGNATURE SCHEMES

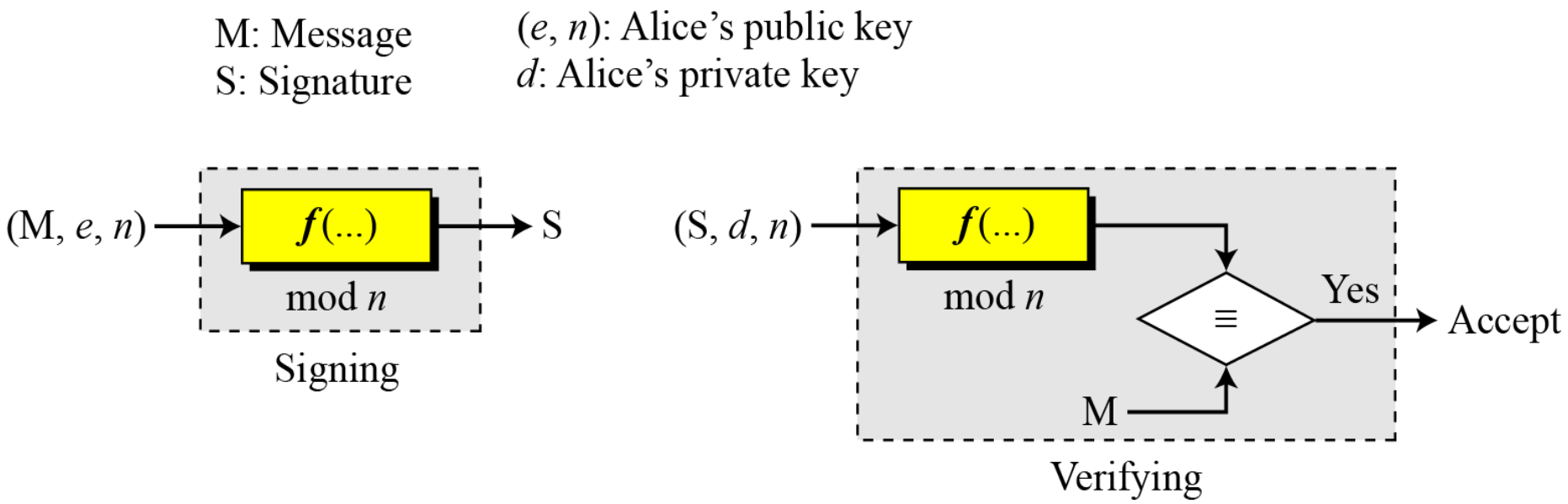
Several digital signature schemes have evolved during the last few decades. Some of them have been implemented.

Topics discussed in this section:

13.5.1 RSA Digital Signature Scheme
13.5.2 ElGamal Digital Signature Scheme
13.5.3 Schnorr Digital Signature Scheme
13.5.4 Digital Signature Standard (DSS)
13.5.5 Elliptic Curve Digital Signature Scheme

13.5.1 RSA Digital Signature Scheme

Figure 13.6 *General idea behind the RSA digital signature scheme*



13.5.1 Key Generation

Key generation in the RSA digital signature scheme is exactly the same as key generation in the RSA

Algorithm 10.2 *RSA Key Generation*

RSA_Key_Generation

{

Select two large primes p and q such that $p \neq q$.

$n \leftarrow p \times q$

$\phi(n) \leftarrow (p - 1) \times (q - 1)$

Select e such that $1 < e < \phi(n)$ and e is coprime to $\phi(n)$

$d \leftarrow e^{-1} \bmod \phi(n)$ // d is inverse of e modulo $\phi(n)$

Public_key $\leftarrow$ (e, n) // To be announced publicly

Private_key $\leftarrow d$ // To be kept secret

return Public_key and Private_key

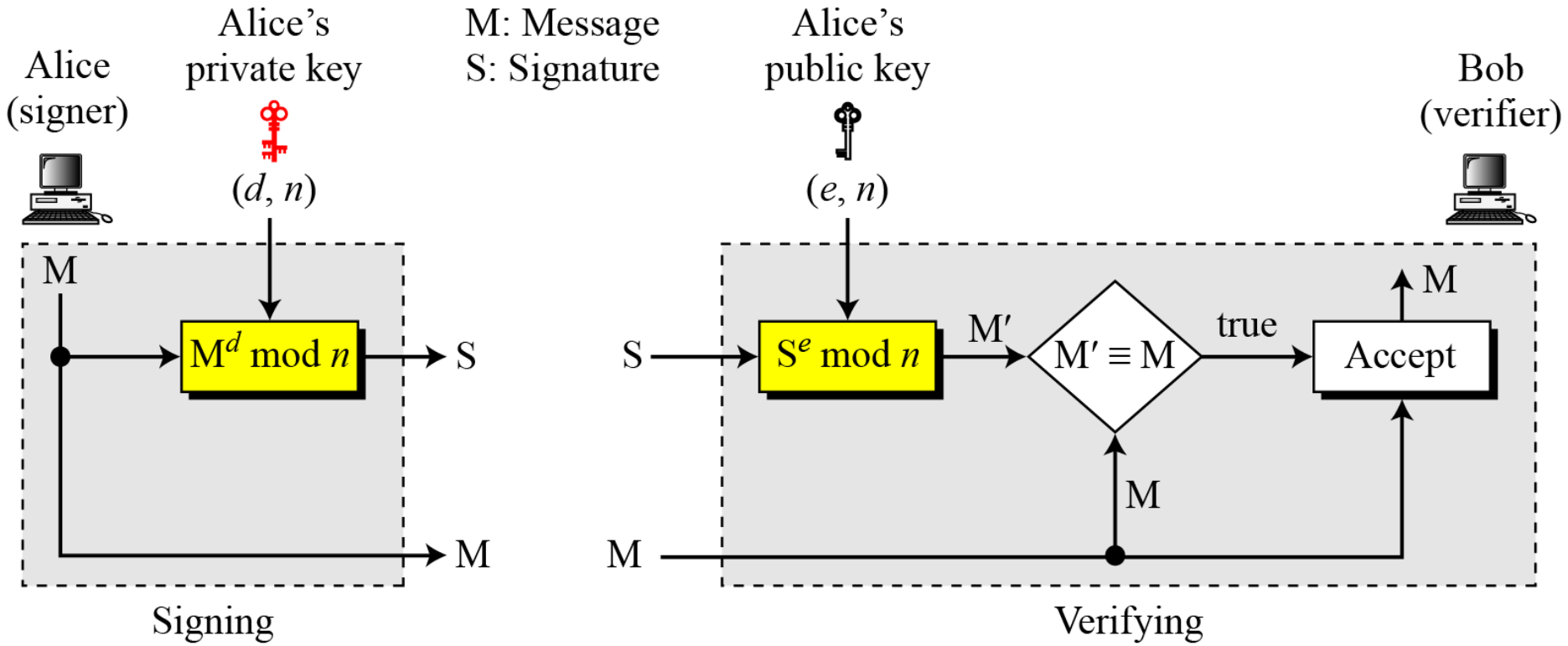
}

Note

In the RSA digital signature scheme, *d* is private; *e* and *n* are public.

Signing and Verifying

Figure 13.7 *RSA digital signature scheme*



13.5.1 Continued

Example 13.1

As a trivial example, suppose that Alice chooses $p = 823$ and $q = 953$, and calculates $n = 784319$. The value of $\phi(n)$ is 782544. Now she chooses $e = 313$ and calculates $d = 160009$. At this point key generation is complete. Now imagine that Alice wants to send a message with the value of M = 19070 to Bob. She uses her private exponent, 160009, to sign the message:

$$\text{M: } 19070 \quad \rightarrow \quad S = (19070^{160009}) \bmod 784319 = 210625 \bmod 784319$$

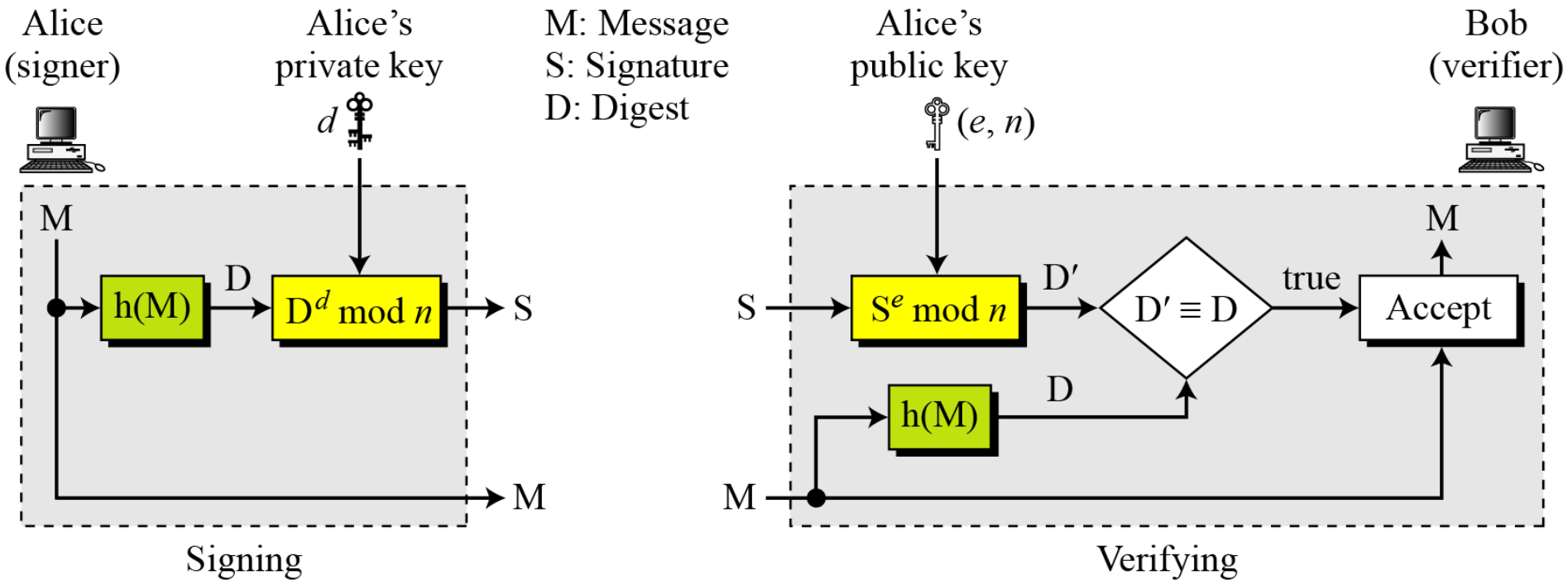
Alice sends the message and the signature to Bob. Bob receives the message and the signature. He calculates

$$M' = 210625^{313} \bmod 784319 = 19070 \bmod 784319 \quad \rightarrow \quad M \equiv M' \bmod n$$

Bob accepts the message because he has verified Alice's signature.

RSA Signature on the Message Digest

Figure 13.8 *The RSA signature on the message digest*

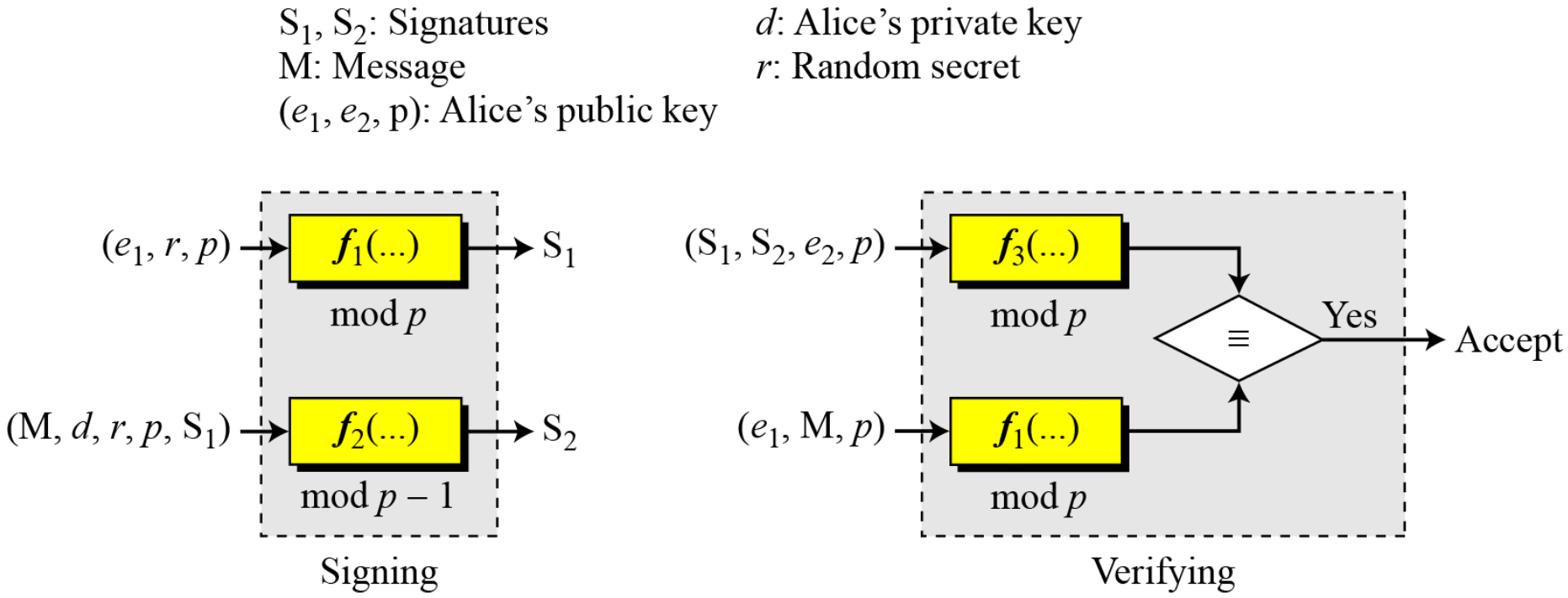


Note

When the digest is signed instead of the message itself, the susceptibility of the RSA digital signature scheme depends on the strength of the hash algorithm.

13.5.2 ElGamal Digital Signature Scheme

Figure 13.9 *General idea behind the ElGamal digital signature scheme*



Key Generation

The key generation procedure here is exactly the same as the one used in the cryptosystem.

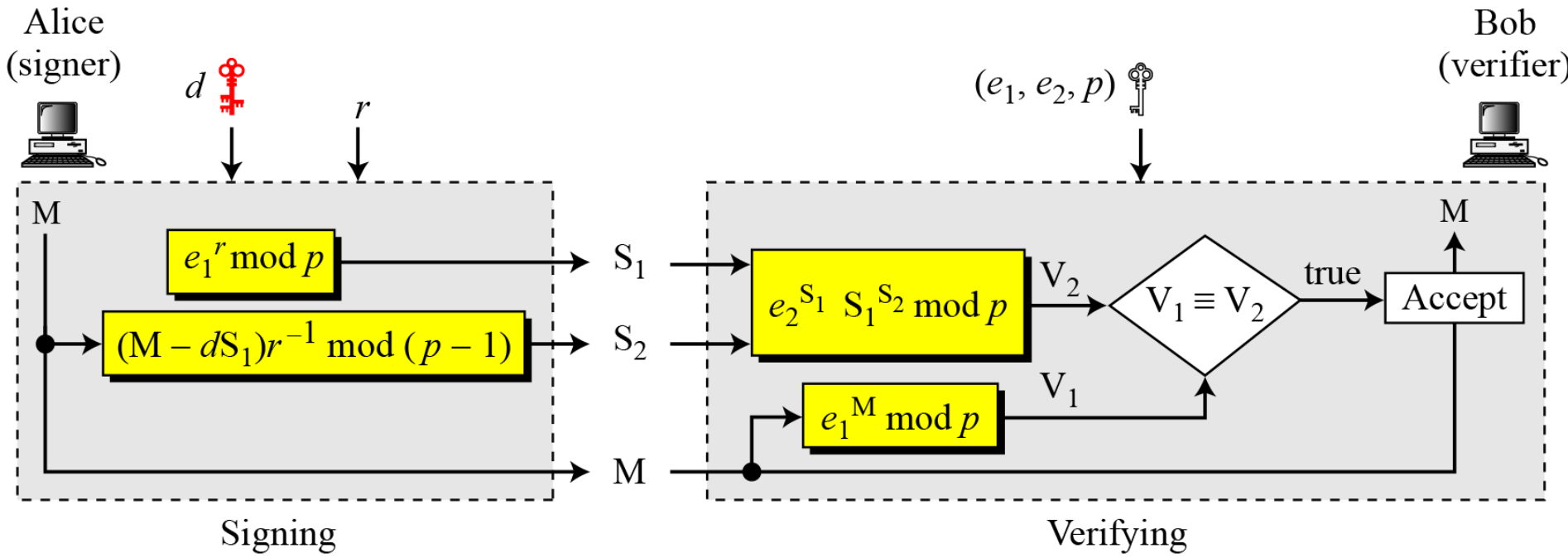
Note

In ElGamal digital signature scheme, (e_1, e_2, p) is Alice's public key; d is her private key.

Verifying and Signing

Figure 13.10 *ElGamal digital signature scheme*

M: Message
S_1, S_2: Signatures
V_1, V_2: Verifications
r: Random secret
d: Alice's private key
(e_1, e_2, p): Alice's public key



13.5.1 Continued

Example 13.2

Here is a trivial example. Alice chooses $p = 3119$, $e_1 = 2$, $_d = 127$ and calculates $e_2 = 2^{127} \bmod 3119 = 1702$. She also chooses r to be 307. She announces e*1*, e*2*, and p publicly; she keeps d secret. The following shows how Alice can sign a message.

$$M = 320$$

$$S_1 = e_1{}^r = 2^{307} = 2083 \bmod 3119$$

$$S_2 = (M - d \times S_1) \times r^{-1} = (320 - 127 \times 2083) \times 307^{-1} = 2105 \bmod 3118$$

Alice sends M, S_1, and S_2 to Bob. Bob uses the public key to calculate V_1 and V_2.

$$V_1 = e_1{}^M = 2^{320} = 3006 \bmod 3119$$

$$V_2 = d^{S_1} \times S_1{}^{S_2} = 1702^{2083} \times 2083^{2105} = 3006 \bmod 3119$$

Example 13.3

Now imagine that Alice wants to send another message, M = 3000, to Ted. She chooses a new *r*, 107. Alice sends M, S_1, and S_2 to Ted. Ted uses the public keys to calculate V_1 and V_2.

$$M = 3000$$

$$S_1 = e_1{}^r = 2^{107} = 2732 \bmod 3119$$

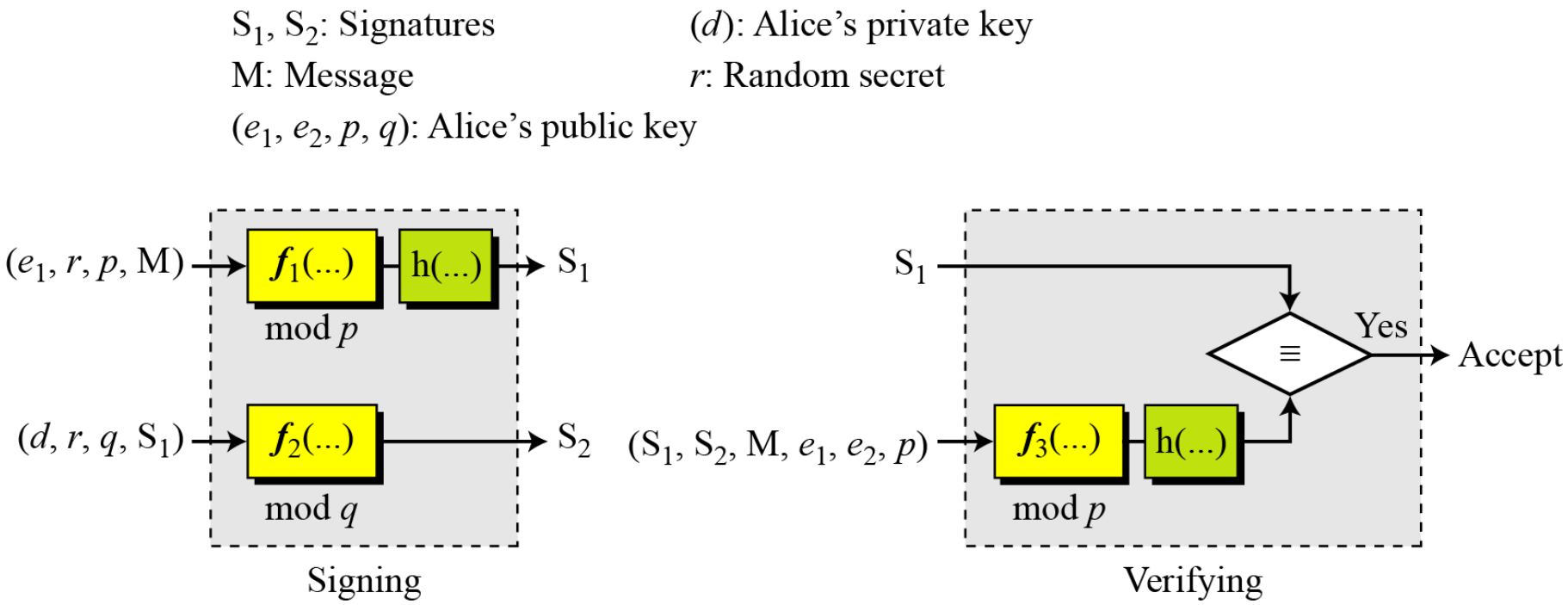
$$S_2 = (M - d \times S_1)\, r^{-1} = (3000 - 127 \times 2083) \times 107^{-1} = 2526 \bmod 3118$$

$$V_1 = e_1{}^M = 2^{3000} = 704 \bmod 3119$$

$$V_2 = d^{S_1} \times S_1{}^S = 1702^{2732} \times 2083^{2526} = 704 \bmod 3119$$

13.5.3 Schnorr Digital Signature Scheme

Figure 13.11 *General idea behind the Schnorr digital signature scheme*



Key Generation

1) Alice selects a prime p, which is usually 1024 bits in length.
2) Alice selects another prime q.
3) Alice chooses e_1 to be the qth root of 1 modulo p.
4) Alice chooses an integer, d, as her private key.
5) Alice calculates $e_2 = e_1^d \bmod p$.
6) Alice's public key is (e_1, e_2, p, q); her private key is (d).

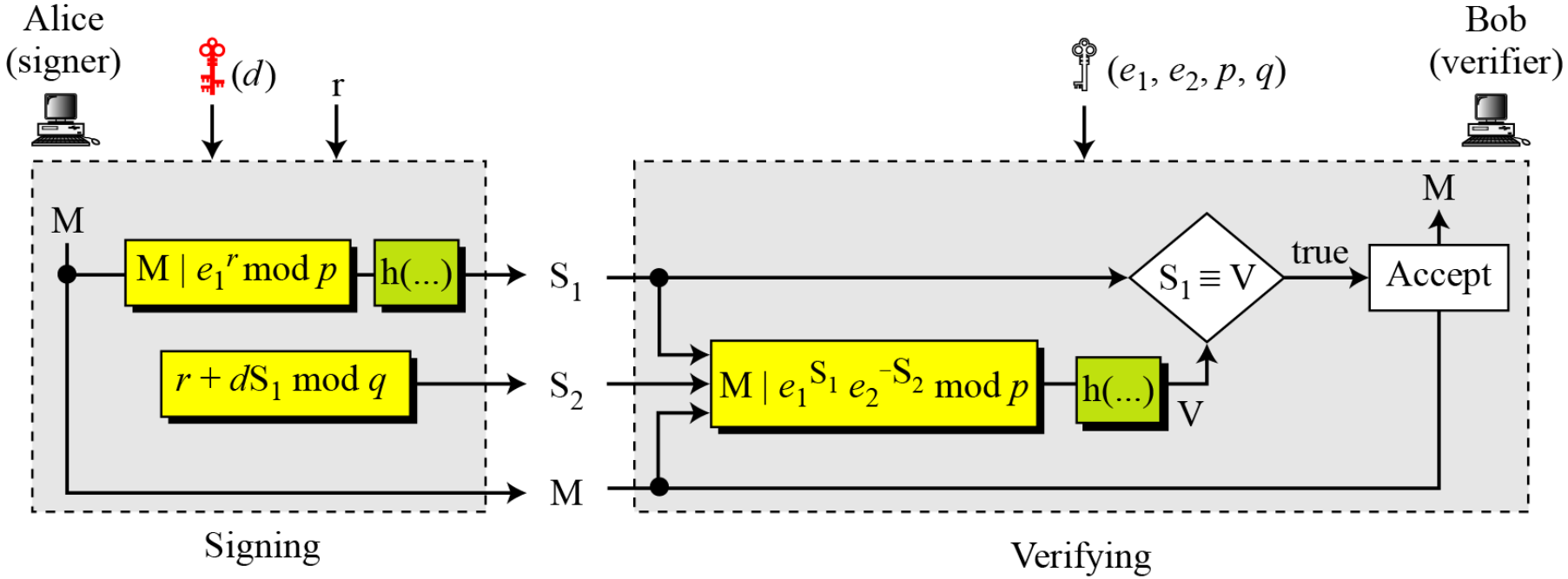
Note

In the Schnorr digital signature scheme, Alice's public key is (e_1, e_2, p, q); her private key (d).

Signing and Verifying

Figure 13.12 *Schnorr digital signature scheme*

M: Message | r: Random secret | | : Concatenation
S_1, S_2: Signatures | (d): Alice's private key | h(...): Hash algorithm
V: Verification | (e_1, e_2, p, q): Alice's public key



Signing

1. Alice chooses a random number r.
2. Alice calculates $S_1 = h(M|e_1^r \bmod p)$.
3. Alice calculates $S_2 = r + d \times S_1 \bmod q$.
4. Alice sends M, S_1, and S_2.

Verifying Message

1. Bob calculates $V = h\,(M \mid e_1^{S2}\, e_2^{-S1} \bmod p)$.
2. If S_1 is congruent to V modulo p, the message is accepted;

13.5.1 Continued

Example 13.4

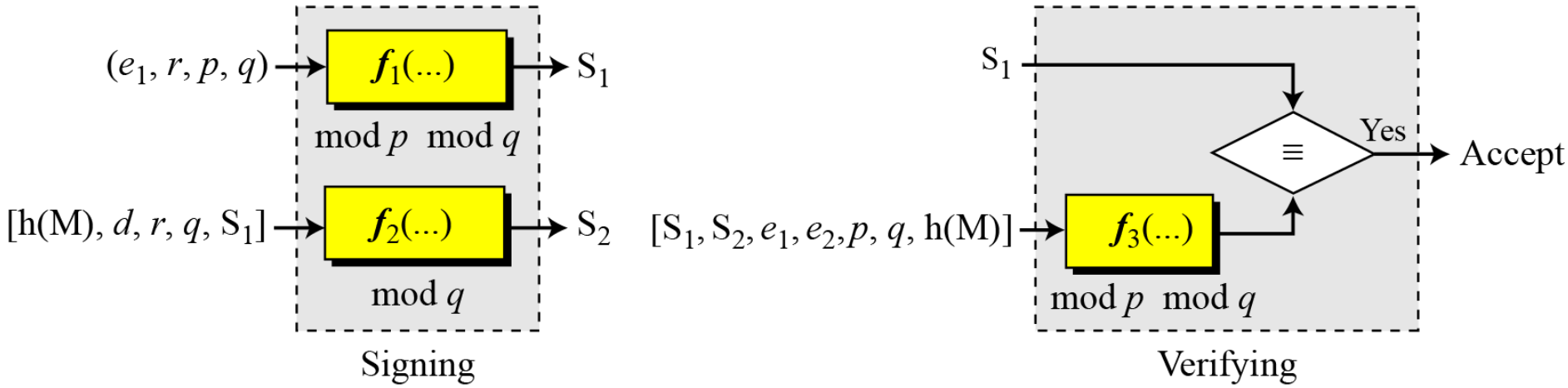
Here is a trivial example. Suppose we choose $q = 103$ and $p = 2267$. Note that $p = 22 \times q + 1$. We choose $e_0 = 2$, which is a primitive in Z_{2267}^*. Then $(p - 1) / q = 22$, so we have $e_1 = 2^{22} \bmod 2267 = 354$. We choose $d = 30$, so $e_2 = 354^{30} \bmod 2267 = 1206$. Alice's private key is now (d); her public key is (e_1, e_2, p, q).

Alice wants to send a message M. She chooses $r = 11$ and calculates $e_2{}^r = 354^{11} = 630 \bmod 2267$. Assume that the message is 1000 and concatenation means 1000630. Also assume that the hash of this value gives the digest h(1000630) = 200. This means S1 = 200. Alice calculates $S2 = r + d \times S_1 \bmod q = 11 + 1026 \times 200 \bmod 103 = 35$. Alice sends the message M =1000, $S_1 = 200$, and $S_2 = 35$. The verification is left as an exercise.

13.5.4 Digital Signature Standard (DSS)

Figure 13.13 *General idea behind DSS scheme*

S_1, S_2: Signatures
M: Message
(e_1, e_2, p, q): Alice's public key
d: Alice's private key
r: Random secret



13.5.4 Continued

Key Generation.

1) *Alice chooses primes p and q.*

2) *Alice uses $<Z_p^*, \times>$ and $<Z_q^*, \times>$.*

3) *Alice creates e_1 to be the qth root of 1 modulo p.*

4) *Alice chooses d and calculates $e_2 = e_1^d$.*

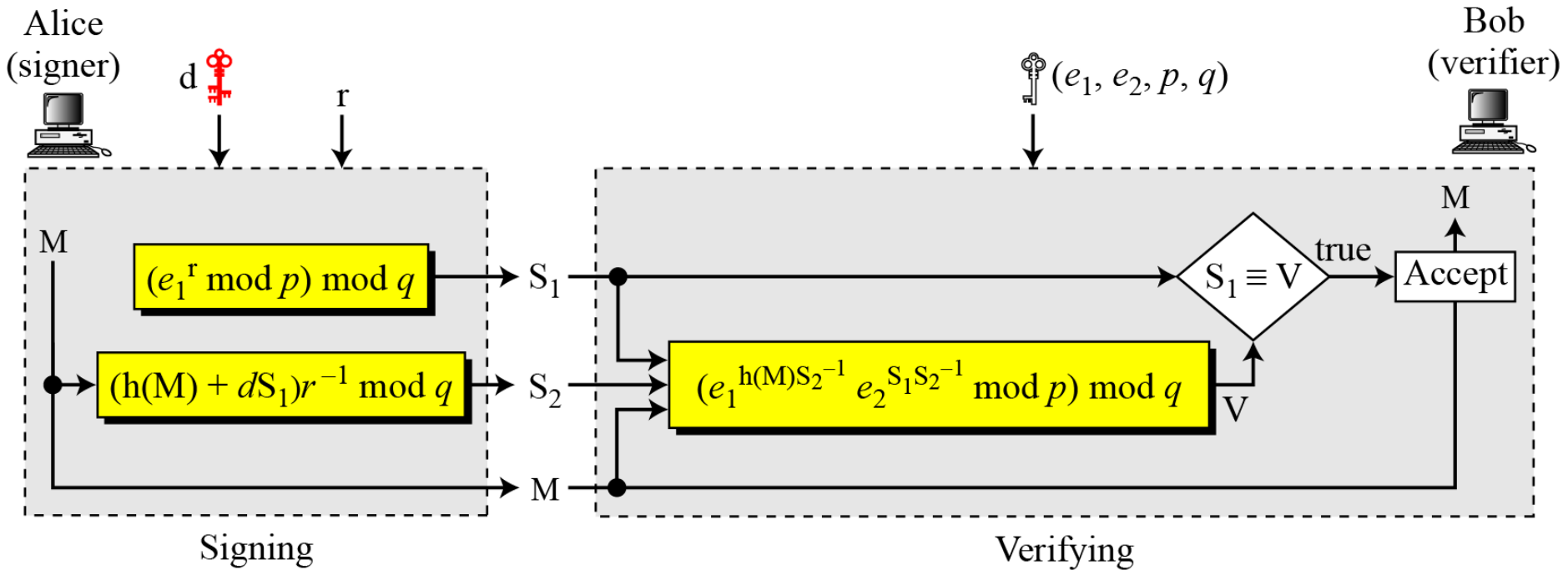
5) *Alice's public key is (e_1, e_2, p, q); her private key is (d).*

13.5.4 Continued

Verifying and Signing

Figure 13.14 *DSS scheme*

M: Message
S_1, S_2: Signatures
V: Verification
r: Random secret
d: Alice's private key
(e_1, e_2, p, q): Alice's public key
h(M): Message digest



13.5.1 Continued

Example 13.5

Alice chooses $q = 101$ and $p = 8081$. Alice *selects* $e_0 = 3$ and calculates $e^1 = e_0^{(p-1)/q} \bmod p = 6968$. Alice chooses $d = 61$ as the private key and calculates $e_2 = e_1^d \bmod p = 2038$. Now Alice can send a message to Bob. Assume that h(M) = 5000 and Alice chooses $r = 61$:

$$h(M) = 5000 \qquad r = 61$$
$$S_1 = (e_1^{\,r} \bmod p) \bmod q = 54$$
$$S_2 = ((h(M) + d\,S_1)\,r^{-1}) \bmod q = 40$$

Alice sends M, S_1, and S_2 to Bob. Bob uses the public keys to calculate V.

$$S_2^{-1} = 48 \bmod 101$$
$$V = [(6968^{5000 \times 48} \times 2038^{54 \times 48}) \bmod 8081] \bmod 101 = 54$$

DSS Versus RSA

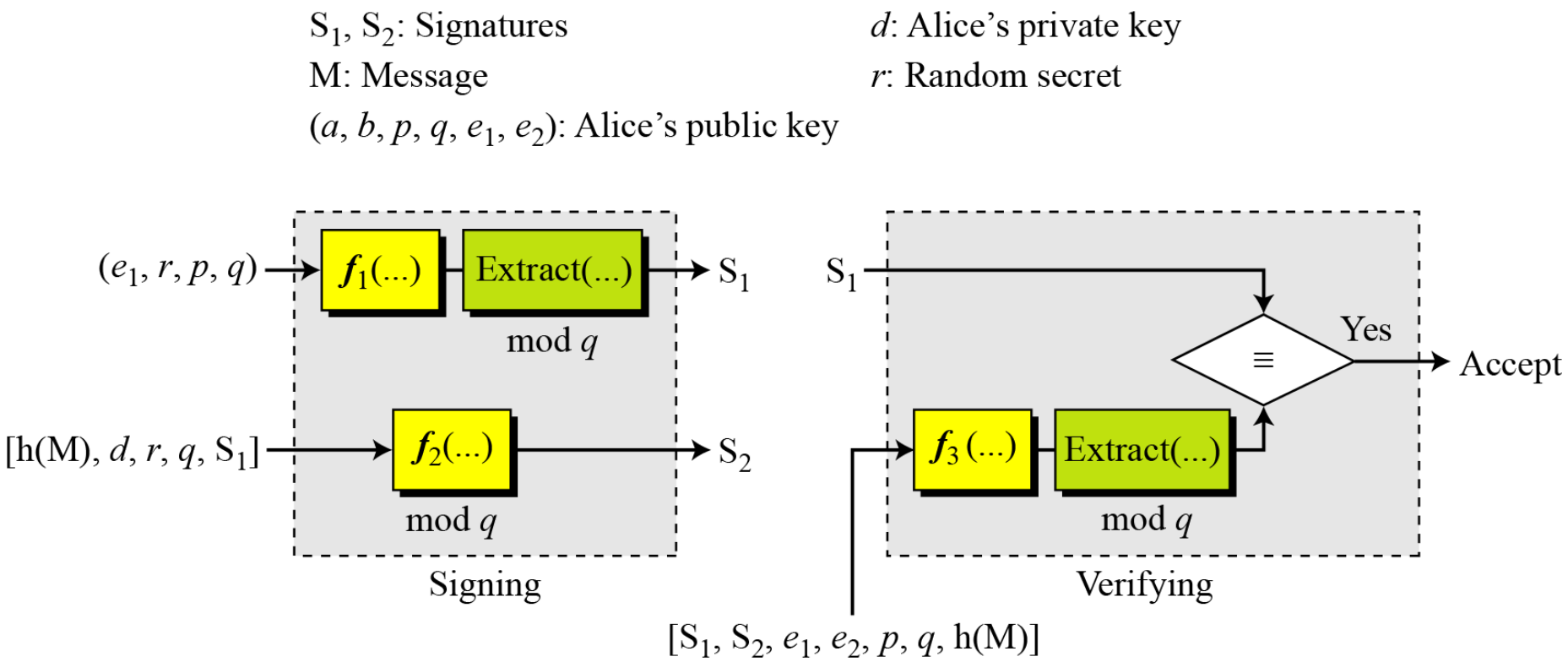
Computation of DSS signatures is faster than computation of RSA signatures when using the same p.

DSS Versus ElGamal

DSS signatures are smaller than ElGamal signatures because q is smaller than p.

13.5.5 Elliptic Curve Digital Signature Scheme

Figure 13.15 *General idea behind the ECDSS scheme*



13.5.5 Continued

Key Generation

Key generation follows these steps:

1) Alice chooses an elliptic curve $E_p(a, b)$.

2) Alice chooses another prime q the private key d.

3) Alice chooses $e_1(\ldots, \ldots)$, a point on the curve.

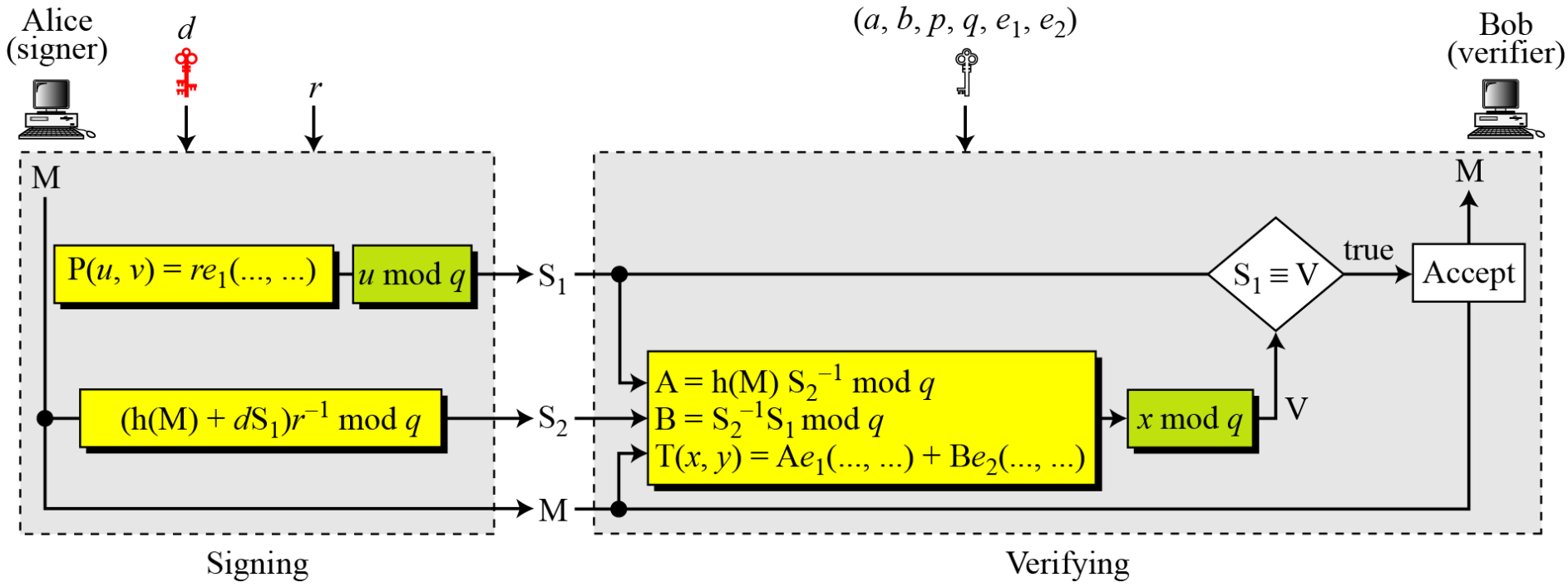
4) Alice calculates $e_2(\ldots, \ldots) = d \times e_1(\ldots, \ldots)$.

5) Alice's public key is (a, b, p, q, e1, e2); her private key is d.

Signing and Verifying

Figure 13.16 *The ECDSS scheme*

M: Message
S_1, S_2: Signatures
V: Verification
r: Random secret
d: Alice's private key
(a, b, p, q, e_1, e_2): Alice's public key
P(u, v), T(x, y): Points on the curve
h(M): Message digest
A, B: Intermediate results



13-6 VARIATIONS AND APPLICATIONS

This section briefly discusses variations and applications for digital signatures.

Topics discussed in this section:

13.6.1 Variations

13.6.2 Applications

13.6.1 Variations

Time Stamped Signatures

Sometimes a signed document needs to be time stamped to prevent it from being replayed by an adversary. This is called time-stamped digital signature scheme.

Blind Signatures

Sometimes we have a document that we want to get signed without revealing the contents of the document to the signer.